Programming Assignment - Real-World Software Applications

Software Engineering Laboratory
Batch IMG

Experiment 4 — 21 August 2026

Student submission details

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Branch: **IMG**

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1 Aim

To design, implement, and test software applications for real-world problems using appropriate programming concepts and software engineering practices.

2 Learning Objectives

- Analyze real-world software problems and design a solution before coding.
- Implement functional menu-driven console applications.
- Apply input validation and error handling.
- Test valid, invalid, boundary, and transaction scenarios.

3 Implemented Applications

3.1 Hospital Appointment Management System

The application stores patient records and supports adding, listing, searching, booking, cancelling, updating, doctor-wise appointment lookup, and appointment report generation. It validates duplicate and missing patient IDs, empty patient and doctor names, invalid dates, and appointment state transitions.

3.2 Restaurant Order Management System

The application maintains a food catalogue and customer order. It supports food item management, order add/remove/update operations, subtotal calculation, discount application, and final bill generation. A 10 percent discount is applied when the subtotal is at least 1000. Quantities cannot be negative, zero when adding, or greater than available stock.

3.3 Vehicle Rental Management System

The application maintains vehicles and their availability state. It supports adding, listing, searching, renting, returning, and filtering vehicles. It rejects duplicate IDs, missing vehicles, empty names, negative prices, renting an already rented vehicle, and returning an already available vehicle.

4 Project Structure

```
src/      Application source code and menu entry point
tests/    Unit and console integration tests
data/     Representative JSON input and verified output files
docs/     Test specification, report, and execution screenshots
```

5 Testing Summary

Tests were executed from the repository root using pytest.

Test area	Result
Hospital domain tests	6 passed
Restaurant domain tests	6 passed
Vehicle domain tests	5 passed
Console menu integration test	1 passed
Lab 4 total	**18 passed**
Full semester suite	**66 passed**

6 Test Case Report

ID	Scenario	Actual result	Status
TC-01	Valid patient record	Patient added successfully	PASS
TC-02	Duplicate patient ID	Duplicate patient ID error	PASS
TC-03	Invalid appointment date	Date validation error	PASS
TC-04	Appointment already booked	Booking rejected	PASS
TC-05	Valid food item	Food item added successfully	PASS
TC-06	Duplicate food ID	Duplicate food ID error	PASS
TC-07	Quantity greater than stock	Stock validation error	PASS
TC-08	Discount boundary bill	Correct subtotal, discount, and total	PASS
TC-09	Rent available vehicle	Vehicle marked rented	PASS
TC-10	Rent already rented vehicle	Rental state error	PASS
TC-11	Return available vehicle	Return state error	PASS
TC-12	Invalid menu or numerical input	Input rejected with error message	PASS

7 Execution Evidence

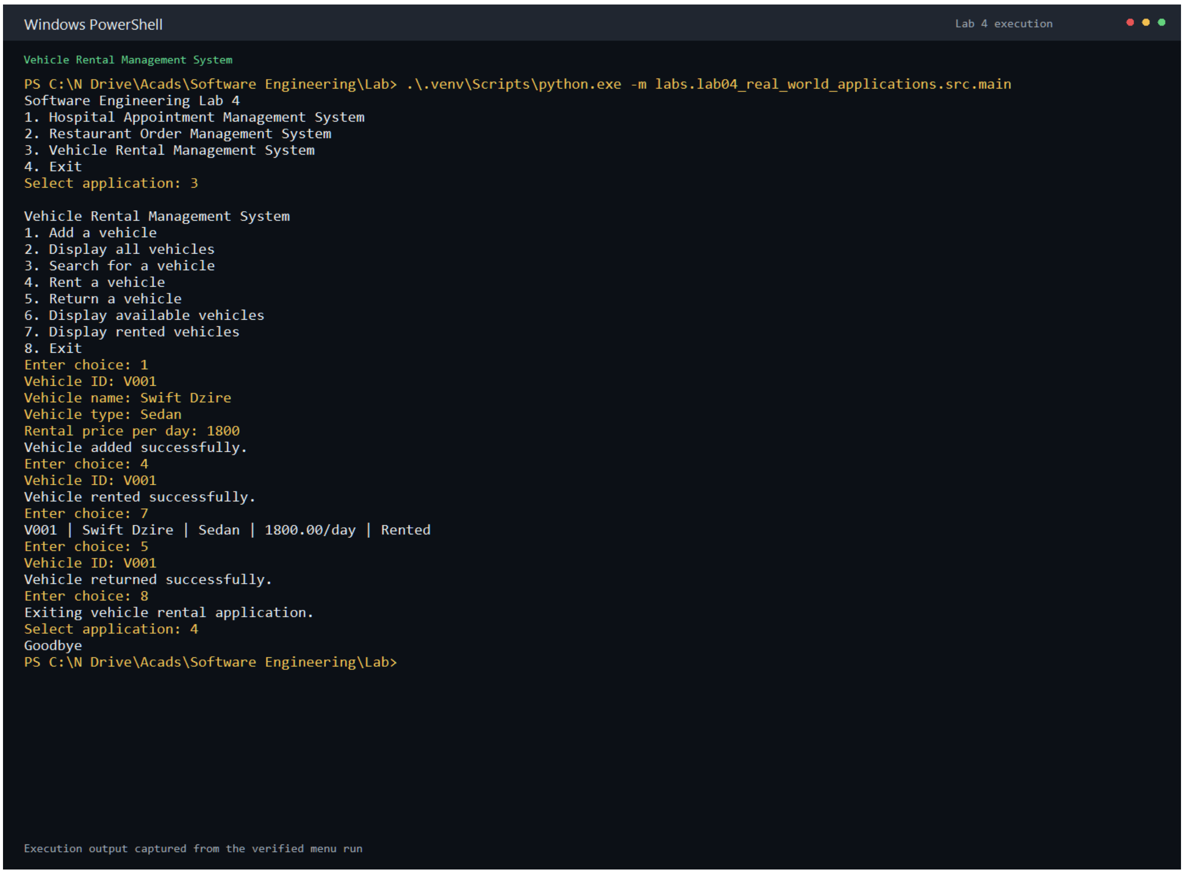
The following pages show the verified output of each menu-driven application. The corresponding plain-text outputs are stored in `data/output/`.

```
Windows PowerShell                                                    Lab 4 execution
Hospital Appointment Management System
PS C:\N Drive\Acads\Software Engineering\Lab> .\.venv\Scripts\python.exe -m labs.lab04_real_world_applications.src.main
Software Engineering Lab 4
1. Hospital Appointment Management System
2. Restaurant Order Management System
3. Vehicle Rental Management System
4. Exit
Select application: 1

Hospital Appointment Management System
1. Add a patient
2. Display all patients
3. Search for a patient
4. Book an appointment
5. Cancel an appointment
6. Update appointment details
7. Display appointments for a doctor
8. Display appointment report
9. Exit
Enter choice: 1
Patient ID: P001
Patient name: Asha Rao
Doctor name: Dr. Mehta
Department: Cardiology
Patient added successfully.
Enter choice: 4
Patient ID: P001
Doctor name: Dr. Mehta
Appointment date (YYYY-MM-DD): 2026-09-01
Appointment booked successfully.
Enter choice: 8
{'total_patients': 1, 'booked': 1, 'cancelled': 0, 'not_booked': 0}
Enter choice: 5
Patient ID: P001
Appointment cancelled successfully.
Enter choice: 9
Exiting hospital application.
Select application: 4
Goodbye
PS C:\N Drive\Acads\Software Engineering\Lab>

Execution output captured from the verified menu run
```

Hospital appointment execution output

```
Windows PowerShell                                                    Lab 4 execution

Restaurant Order Management System

PS C:\N Drive\Acads\Software Engineering\Lab> .\.venv\Scripts\python.exe -m labs.lab04_real_world_applications.src.main
Software Engineering Lab 4
1. Hospital Appointment Management System
2. Restaurant Order Management System
3. Vehicle Rental Management System
4. Exit
Select application: 2

Restaurant Order Management System
1. Add a food item
2. Display all food items
3. Search for a food item
4. Add a food item to an order
5. Remove a food item from an order
6. Update food quantity
7. Display the customer order
8. Calculate the bill
9. Apply discount
10. Generate final bill
11. Exit
Enter choice: 1
Food ID: F001
Food name: Veg Thali
Category: Main Course
Price: 250
Available quantity: 10
Food item added successfully.
Enter choice: 4
Food ID: F001
Quantity: 5
Food item added to order.
Enter choice: 10
Subtotal: 1250.00
Discount: 125.00
Final total: 1125.00
Enter choice: 11
Exiting restaurant application.
Select application: 4
Goodbye
PS C:\N Drive\Acads\Software Engineering\Lab>


Execution output captured from the verified menu run
```

Restaurant order and bill execution output



Vehicle rental execution output

8 Submission Contents

- Source code: `src/`
- Unit and integration tests: `tests/`
- Input data: `data/input/`
- Output files: `data/output/`
- Execution screenshots: `docs/screenshots/`
- Test specification: `docs/test_specification.md`
- Editable test report: `docs/test_execution_report.md`

9 Conclusion

All three requested real-world applications were implemented as menu-driven console programs. The applications include validation and error handling for the required invalid scenarios, and the automated test suite confirms the main business operations and menu behavior.